

Welcome to

Network Attacks II – Hands-on & Defence

Saint Cyber – Cyber Akademiet, 2026

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Slides are available as PDF, https://codeberg.org/kramse
academy-2026-day1-network-attacks-II.tex in the repo security-courses

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Goals for today



- Demonstrate network-attack techniques in an authorised lab
- Observe the effect of flooding on a target
- Recommend defensive measures: patching, monitoring, rate/connection limits, filtering, CDN and DDoS protection

Plan for today



- Lab environment and authorisation
- hping3 – crafting floods
- Scapy – packet crafting, task and challenge
- Defences and mitigations
- Red/blue exercise

Time schedule



- **FIXME:** 1) hping3 flooding – XX min
- **FIXME:** 2) Scapy + task – XX min
- **FIXME:** 3) Defences – XX min
- **FIXME:** 4) Red/blue exercise – XX min

Lab environment & authorisation



- **FIXME:** describe the lab, target VMs, and the "only in the lab" rule

hping3 – crafting floods



- **FIXME:** example commands for SYN/UDP/ICMP floods – test before class

Observing the impact



- **FIXME:** Wireshark, nload, netstat / TCPview – what to look for

Scapy – packet crafting



- **FIXME:** the basics the cadets need (IP()/TCP(), send/sendp, sniff)

Scapy – task



- **FIXME:** write the task (reuse your existing Scapy exercise)

Scapy – challenge



- **FIXME:** the spoofed-source / multi-port challenge

Defences: patching & monitoring



- **FIXME:** keep firmware/OS/software patched; detect early

Defences: rate limits & filtering



- **FIXME:** rate/connection limits, ingress filtering

Defences: CDN & DDoS protection



- **FIXME:** what a CDN does; scrubbing / DDoS protection services

Red/blue exercise



- **FIXME:** the attack-and-defend exercise for the cadets

Summary & references



- **FIXME:** recap attacks + the matching defences
- **FIXME:** add references / tool links